

The background of the entire image is a composite. The top half features a large, waving American flag. The bottom half shows the Statue of Liberty in green, holding a torch and a tablet, with the New York City skyline at night visible in the background. The text is overlaid on the left side of the image.

A SQL PORTFOLIO PROJECT

Who gets a mortgage in New York?

1.75 million real applications, modelled in PostgreSQL and runnable in
your browser.

1,755,419 applications · 2022 to 2025 · New York State

Samuel Babajide

swipe →

THE QUESTION

Who gets mortgage credit, on what terms, and does the answer change once you hold income, product, neighbourhood and lender constant?

Four years of the Home Mortgage Disclosure Act register for New York State. **Every one of the 1,755,419 published applications is accounted for.**

What four years looks like

1,596,676

applications

purchased loans excluded: nobody applied for those

996,548

approved, a 62.4% approval rate

the lender said yes and the applicant took it

\$371bn

actually disbursed

money that left a lender and reached a borrower

Approval fell 2.8 points in 2023 and never came back. Volume recovered to 79% of 2022. The willingness to say yes did not.

FINDING 01

30.6

percentage points. The median gap between the strictest and the most generous lender **in the same census tract (location)**.

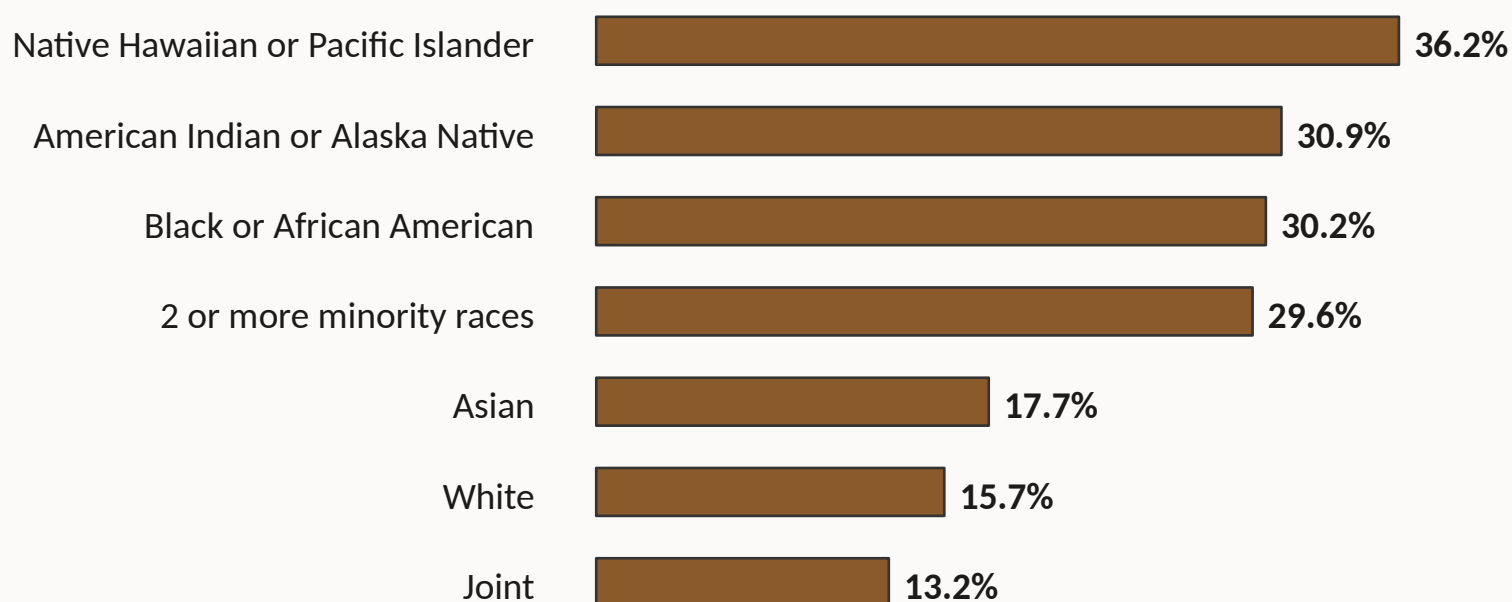
A census tract is a small neighbourhood, roughly 1,200 to 8,000 people. Measured across 532 tracts where four or more lenders each decided 30 or more applications. At the extreme the gap reaches 97.3 points.

Two people on the same street can get opposite answers, depending on which lender they walk into. That matters more than anything else measured in this project.

FINDING 02

Income does not explain the gap

Denial rate in 2025 among applicants earning over \$200,000 a year, by reported race:



Group sizes differ enormously, from 191,212 White applicants to 564 Native Hawaiian or Other Pacific Islander. The three smallest groups move on small numbers.

The gap is between 13.6 and 19.3 points in **every** income band. A Black applicant earning over \$200,000 is denied more often than a White applicant earning between \$50,000 and \$100,000.

FINDING 03

Denial is not the only way out of the funnel

Share of applications ending in "file closed for incompleteness", among lenders with 20,000 or more applications:

lowest lender

0.2%

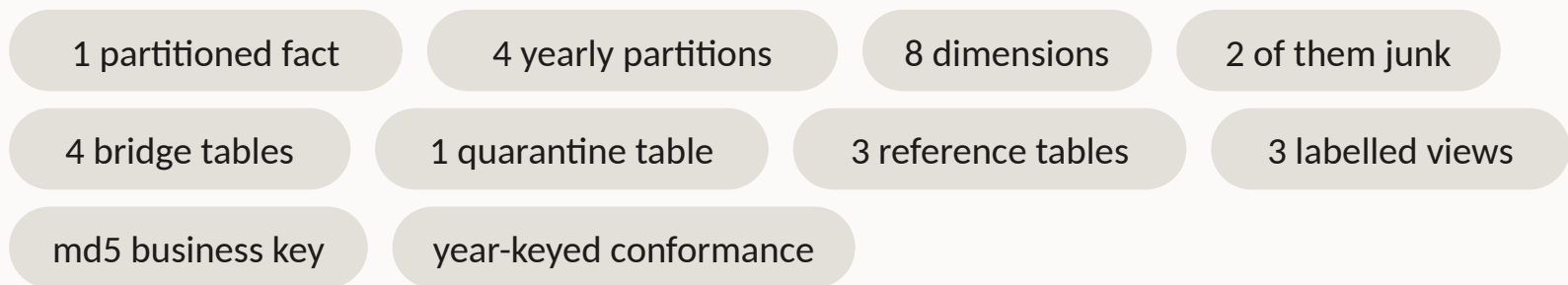
highest lender

28.5%

A factor of 142 between two national banks, in the same state, in the same years, under the same rules. An applicant can be turned away without ever being denied, and a review that counts only denials will never see it.

THE MODEL

99 flat columns into 17 tables



175 columns, 100% documented. The published data dictionary is generated from pg_catalog at build time, so it cannot describe a column that no longer exists.

HMDA strips every loan identifier before publication, so the business key is an md5 of the source row. Rebuild the model on another machine and you get the same keys.

WHAT THIS CANNOT SAY

HMDA contains no credit score. So none of this is causal, and none of it establishes discrimination.

What the data does support is narrower and still substantial: the disparity is not explained by income, by product, by neighbourhood or by lender choice, because it persists after holding each of those constant.

A build that never admits to a gap is not a build anyone should trust.

DON'T TAKE MY WORD FOR IT

Run the queries yourself

1.75M

rows, queried in your browser

no install, no account, no server, and no data leaves your machine

<1s

typical query time

DuckDB compiled to WebAssembly, reading Parquet over range requests

Every figure in this carousel is one click from the SQL that produced it.

**[github.com/SamuelBabajide/
us-home-mortgage-disclosure-act-new-york](https://github.com/SamuelBabajide/us-home-mortgage-disclosure-act-new-york)**

YOUR TURN

What would you ask this data?

Open the playground, change a query, press Ctrl + Enter. If you find something I missed, I would like to know.

BUILT WITH

PostgreSQL 18

dimensional modelling

data quality assertions

query optimisation

Python

DuckDB WASM

Parquet

GitHub Pages